Pilot 5: E4M- Smart Electronics Manufacturing as a Service (electronics manufacturing sector)

MaaSAI 3rd Plenary Meeting – 4th –5th February, 2026 – Tampere, Finland

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Co-funded by the European Union

Outline

PILOT USE CASE DESCRIPTION

BUSINESS PROCESS

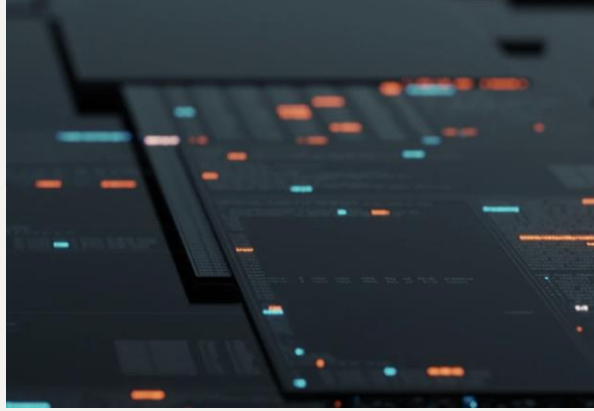
REQUIRMENTS

KPIs

DATA

ACTION POINTS FOR UPCOMING MONTHS

Pilot 5: E4M- Smart Electronics Manufacturing as a Service



Partners involved

Context

Industrial Partner: E4M

TECH Partner: UNINOVA

E4M - Engenharia Mecatrónica Lda

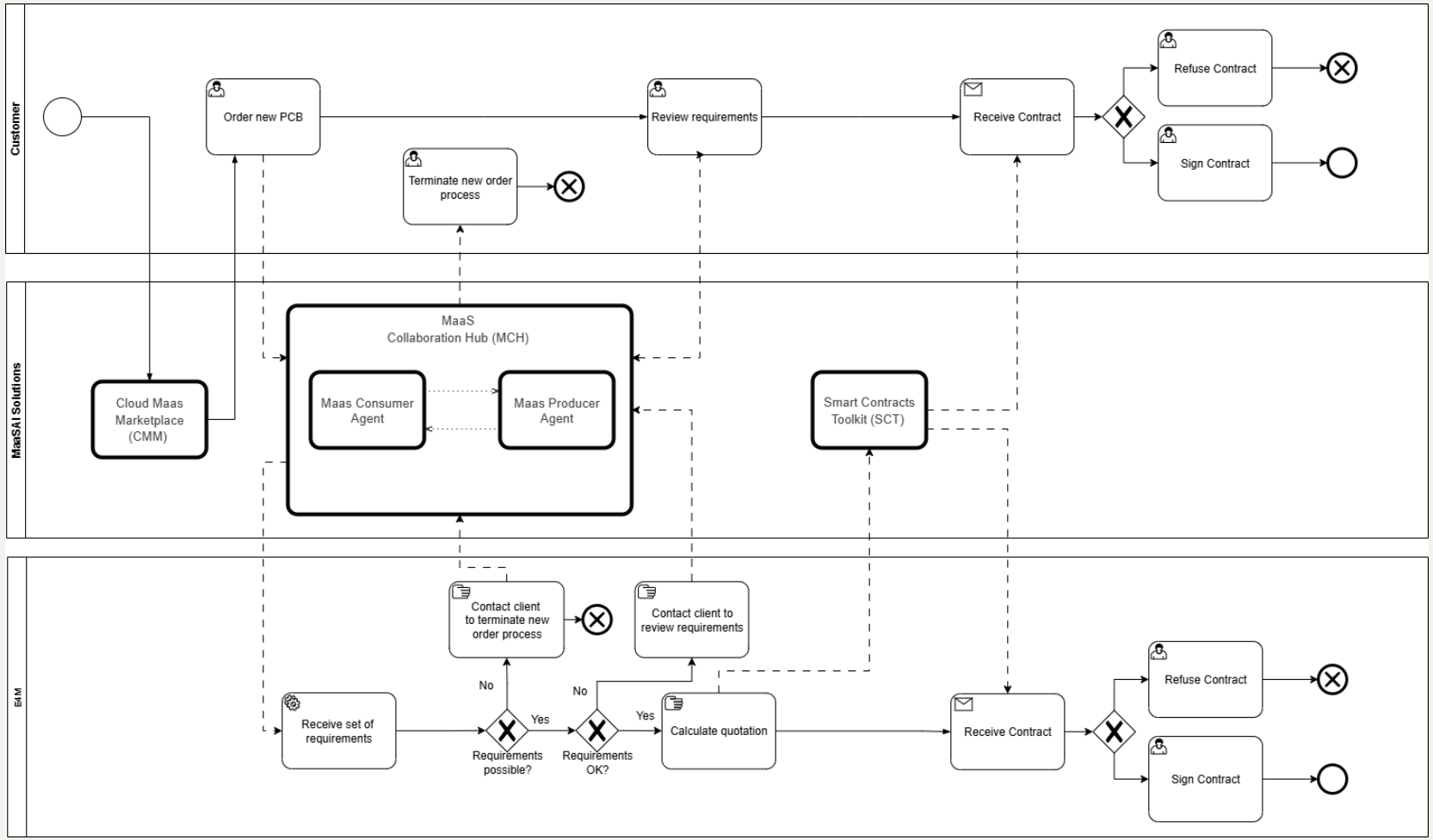
- Core Services:
  - Develop complete or integrated technological solutions for multiples sectors, based on clients needs, in:
    - Process Automation
    - Retrofitting Industrial Machines
    - Mechanical & Electronic Design
    - Electronics Manufacturing & 3D Prototyping
    - End-to-End Project Support
  - PCB development and prototyping and posterior setup and deployment.

Business Process

- **BP01:** Customer Initial Interaction & Commercial Information Exchange
- **BP02:** Development Process Monitoring, Optimisation and Information Exchange
- **BP03:** Maintenance Process Monitoring and Proactivity
- **BP04:** End-of-Life Policy Automation

Business Process 1

- Customer Initial Interaction & Commercial Information Exchange



Business Process 1

- Requirements

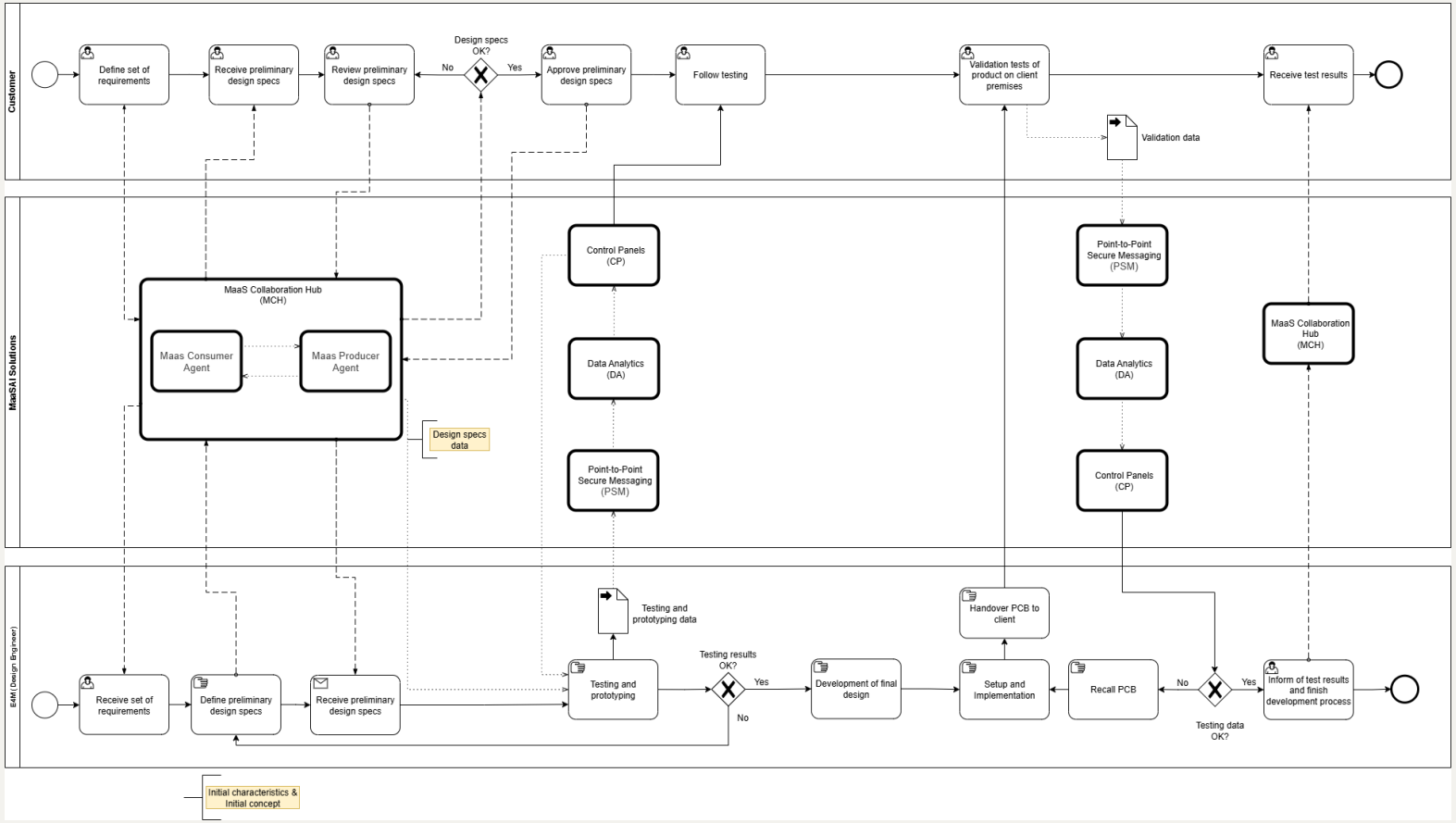
Improvement	Solution
Enable secure communciation with consumer	CMM, MCA, MPA, MCH, PSM
Enable anonymous consumer/provider interaction	CMM
Enable fast and reliable communication with consumer	CMM, MCH
Enable fast communication between the provider and consumer during requirement and design phases	MCH, PSM
Enable fast and reliable business agreements.	SCT, MSB
Enable secure contracts history registry	SCT, MSB

Business Process 1

- KPI1 P5 BP01 - Manufacturing Time Availability Gain
  - Improved and more efficient client interaction process -> More availability

Business Process 2

- Development Process Monitoring, Optimisation and Information Exchange



Business Process 2

- Requirements

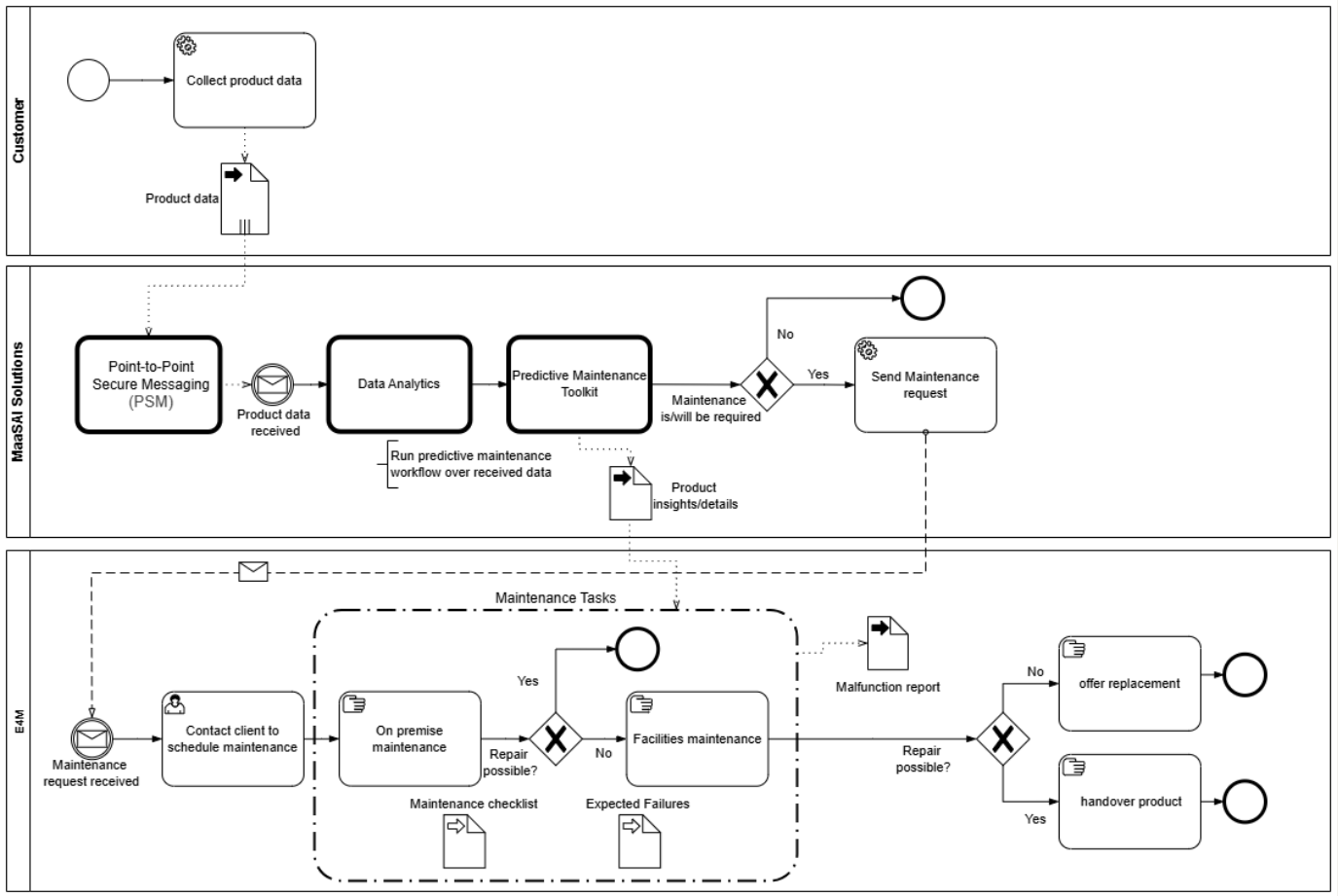
Improvement	Solution
Enable secure communication with consumer	MCH, PSM
Enable fast and reliable interaction with consumer in development phase	CP
Enable the exchange of development, testing and validation statuses between the provider and consumer	CP
Enable the transparency of product development between Provider and Consumer	DA
Enable production and testing data processing and analysis, prediction and optimization of processes	DA

Business Process 2

- KPI1 P5 BP02 - Manufacturing Time Availability Gain
  - Improved and more efficient development/testing/validation process -> More availability
- KPI2 P5 BP02 - Production Flow Efficiency Improvement
  - Improved and more efficient development process -> more production procedures
- KPI3 P5 BP02 - Waste Avoidance Through Remanufacturing
  - Improved and more efficient development process -> more new retrofitting PCBs

Business Process 3

- Maintenance Process Monitoring and Proactivity



Business Process 3

- Requirements

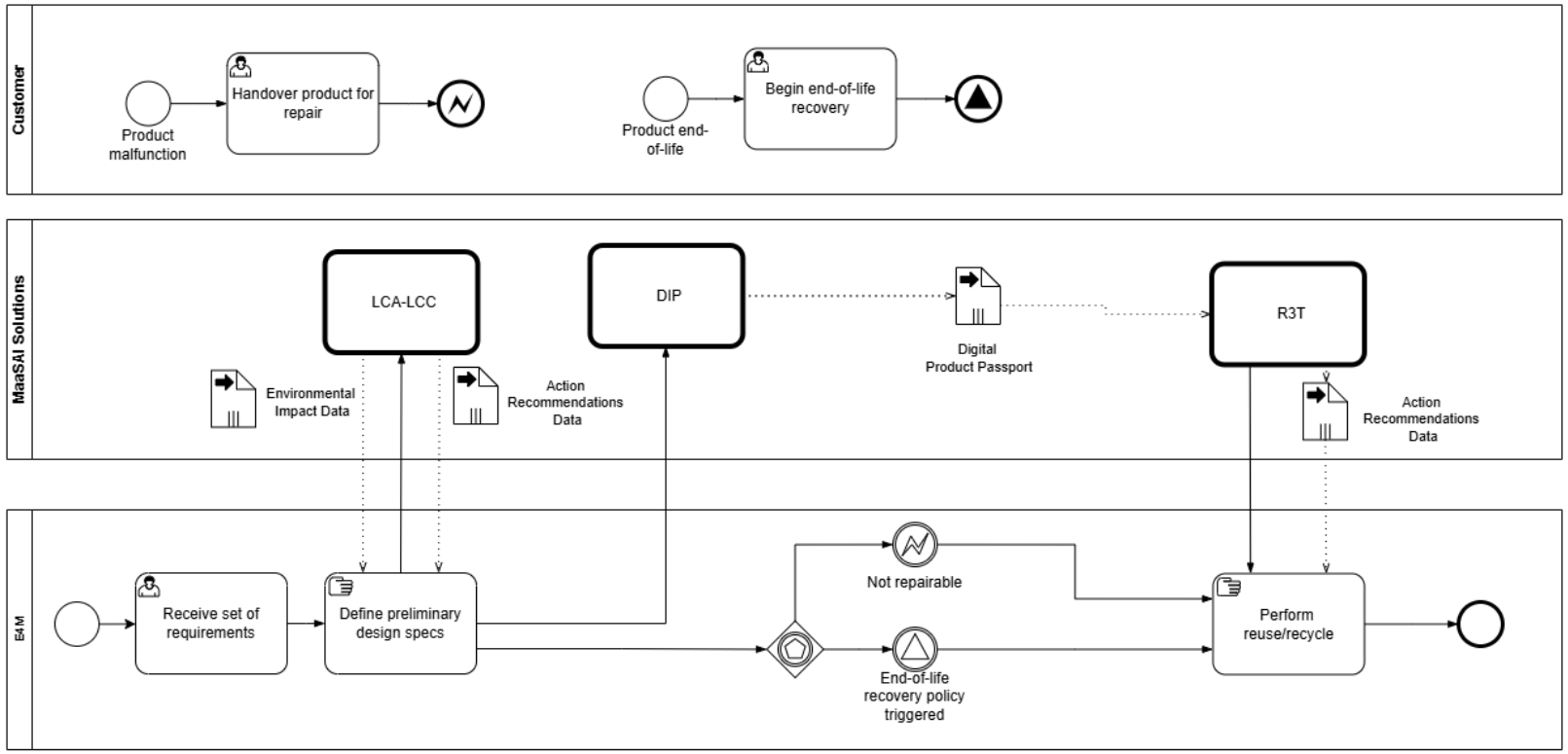
Improvement	Solution
Reduce maintenance related downtimes	DA, PMT, PSM

Business Process 3

- KPI1 P5 BP03 - MTBF Improvement after Datalog Monitoring
  - Better Maintenance process (PCB monitoring) -> increased MTBF
- KPI2 P5 BP03 - MTBF Improvement from Engineering Solutions
  - Better Maintenance process (monitoring inputs to new solutions) -> increased MTBF
- KPI3 P5 BP03 - Repair Time Improvement with MaaS Solutions
  - Better Maintenance process (Predictive) -> less repair time

Business Process 4

- End-of-Life Policy Automation



Business Process 4

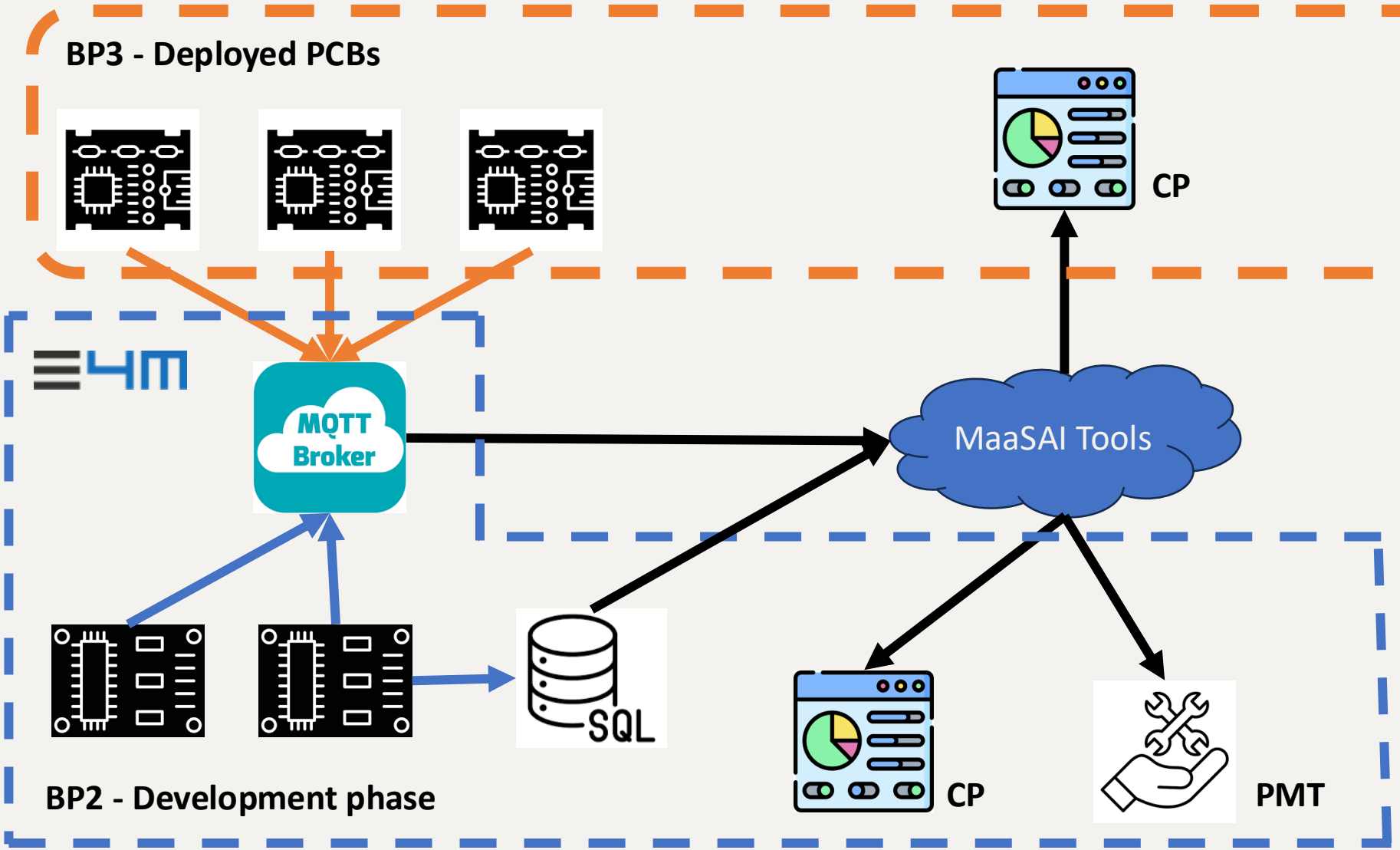
- Requirements

Improvement	Solution
Enable the creation of product design scenarios to reduce environmental impact	LCC, R3T
Enable the creation of production scenarios to reduce environmental impact	LCC, MPP, SCS, DI
Enable fast discovery of E4M products information	DIP

Business Process 4

- KPI1 P5 BP04 - GHG Savings from Engineering Device
  - Use of LCA-LCC on new PCB -> GHG savings
- KPI2 P5 BP04 - Waste Avoidance Through Remanufacturing
  - Use of R3T on new PCB -> recycle/repair/reuse
- KPI3 P5 BP04 - Decrease in energy consumption of the equipments remanufactured
  - Use of R3T/LCA-LCC on new PCB -> decrease energy
- KPI4 P5 BP04 - Product Remanufacturing Rate
  - Use of R3T on new PCB -> repair/reuse

Data


BP3 - Deployed PCBs
Consumer
CP
MQTT Broker
MaaSAI Tools
SQL
CP
PMT
BP2 - Development phase

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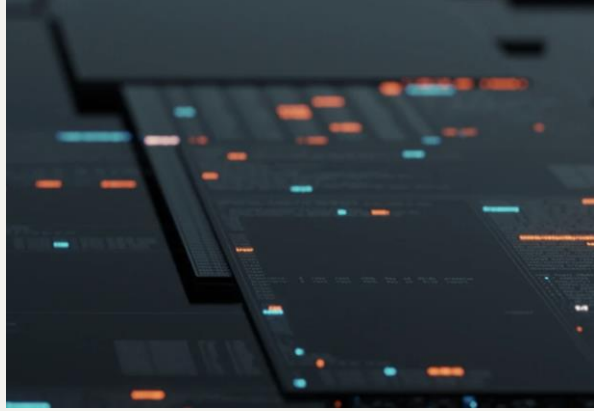

Completed/Finalized Action points:

- Business process scenarios
- Requirements list
- KPIs

Action points for upcoming period:

- Specific use case definition (PCB)
- Baseline measurement for KPIs

MAASAI

Thank you

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